the input signal is processed with an electronic low-pass filter to remove all frequencies above the Nyquist frequency (one-half the sampling rate). This is done to prevent aliasing during sampling, and is correspondingly called an **antialias filter**. On the other end, the digitized signal is passed through a digital-to-analog converter and another low-pass filter set to the Nyquist frequency. This output filter is called a **reconstruction filter**, and may include the previously described zeroth-order-hold frequency boost. Unfortunately, there is a serious problem with this simple model: the limitations of electronic filters can be as bad as the problems they are trying to prevent.

If your main interest is in software, you are probably thinking that you don't need to read this section. *Wrong*! Even if you have vowed never to touch an oscilloscope, an understanding of the properties of analog filters is important for successful DSP. First, the characteristics of every digitized signal you encounter will depend on what type of antialias filter was used when it was acquired. If you don't understand the nature of the antialias filter, you cannot understand the nature of the digital signal. Second, the future of DSP is to replace *hardware* with *software*. For example, the *multirate* techniques presented later in this chapter reduce the need for antialias and reconstruction filters by fancy software tricks. If you don't understand the hardware, you cannot design software to replace it. Third, much of DSP is related to digital filter design. A common strategy is to start with an equivalent *analog filter*, and convert it into software. Later chapters assume you have a basic knowledge of analog filter techniques.

Three types of analog filters are commonly used: **Chebyshev**, **Butterworth**, and **Bessel** (also called a Thompson filter). Each of these is designed to optimize a different performance parameter. The complexity of each filter can be adjusted by selecting the number of **poles** and **zeros**, mathematical terms that will be discussed in later chapters. The more poles in a filter, the more electronics it requires, and the better it performs. Each of these names describe what the filter *does*, not a particular arrangement of resistors and capacitors. For example, a six pole Bessel filter can be implemented by many different types of circuits, all of which have the same overall characteristics. For DSP purposes, the characteristics of these filters are more important than how they are constructed. Nevertheless, we will start with a short segment on the electronic design of these filters to provide an overall framework.

Figure 3-8 shows a common building block for analog filter design, the modified Sallen-Key circuit. This is named after the authors of a 1950s paper describing the technique. The circuit shown is a two pole low-pass filter that can be configured as any of the three basic types. Table 3-1 provides the necessary information to select the appropriate resistors and capacitors. For example, to design a 1 kHz, 2 pole Butterworth filter, Table 3-1 provides the parameters: $k_1 = 0.1592$ and $k_2 = 0.586$. Arbitrarily selecting $R_1 = 10K$ and $C = 0.01uF$ (common values for op amp circuits), R and R_f can be calculated as 15.95K and 5.86K, respectively. Rounding these last two values to the nearest 1% standard resistors, results in $R = 15.8K$ and $R_f = 5.90K$ All of the components should be 1% precision or better.